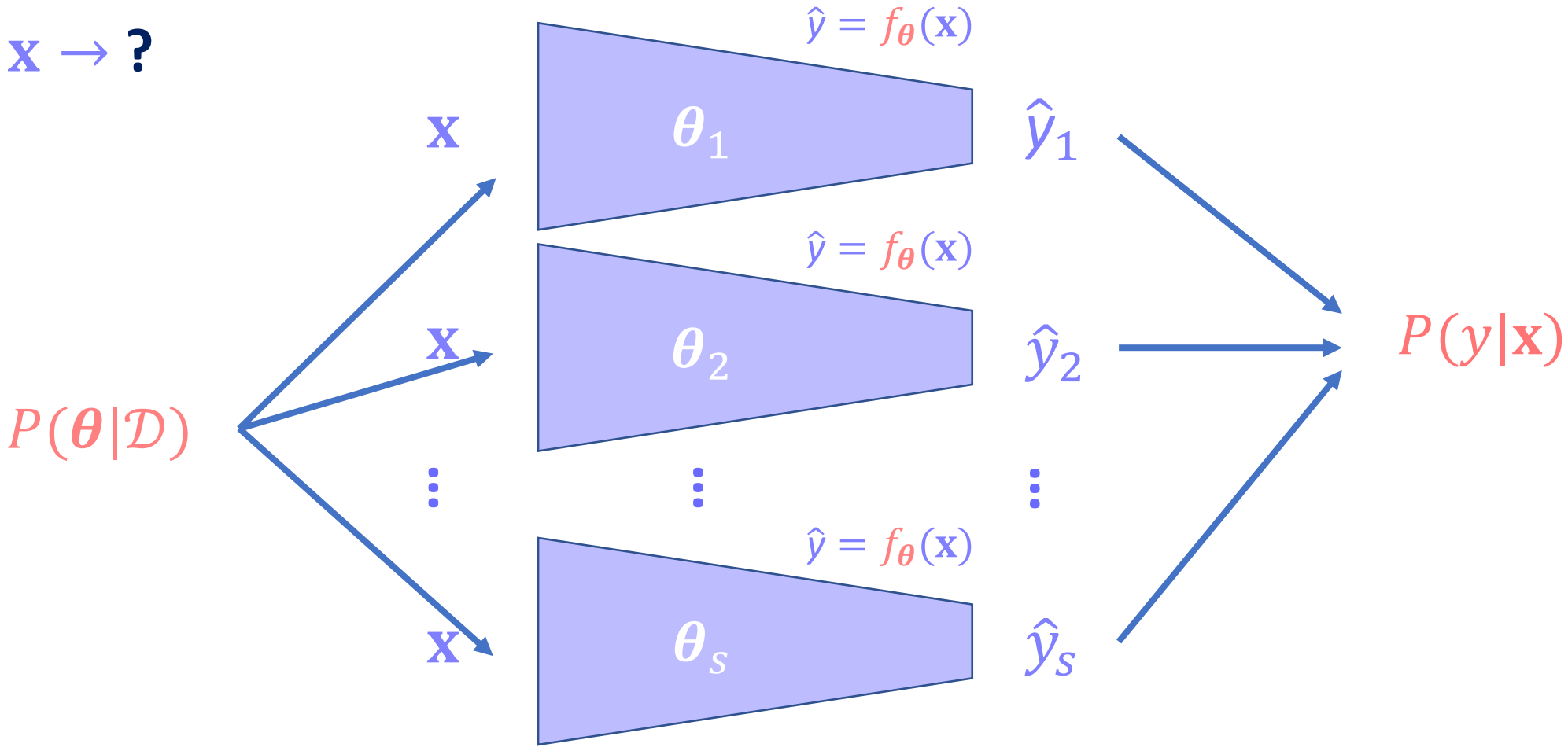
What about non-Bayesian methods?



Simple&Effective Techniques: Ensemble Methods

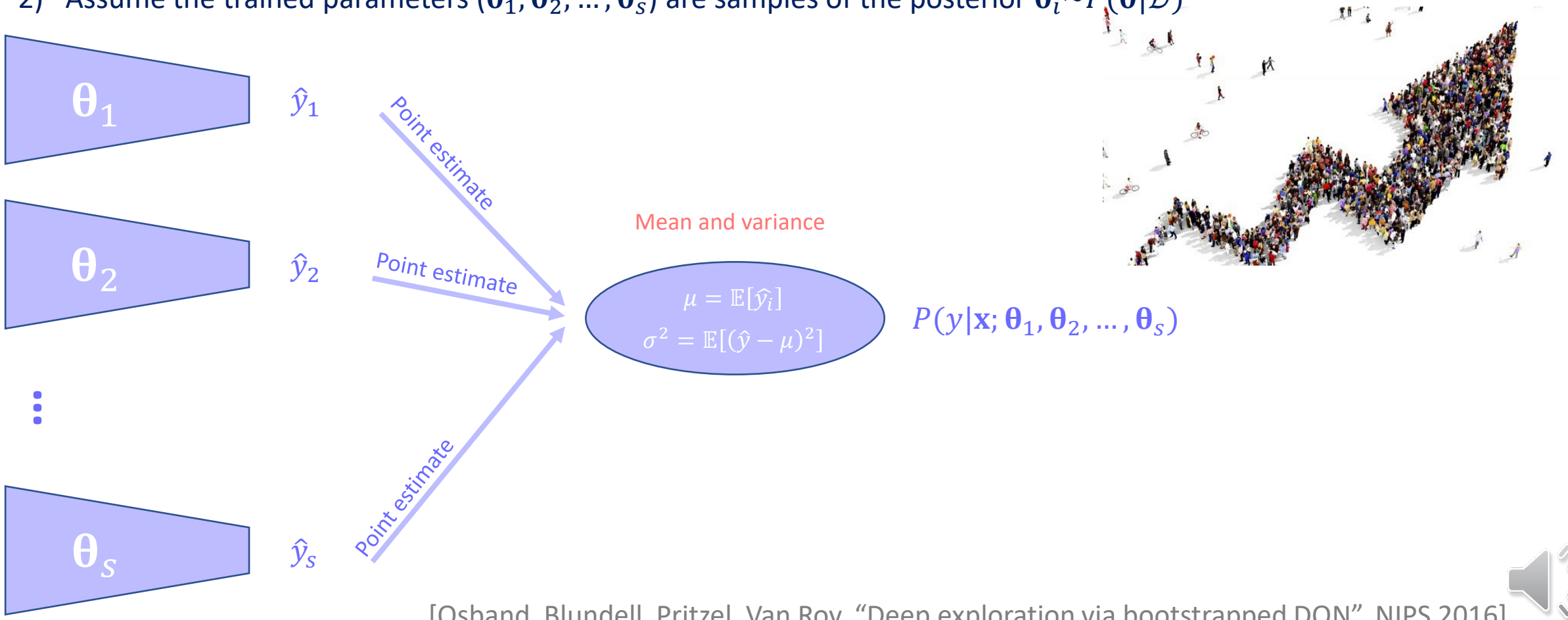
Ensemble-methods: intuition



Simple&Effective Techniques: Ensemble Methods

Ensemble-methods: procedure

1) Train s independent networks using different network initializations and bootstrapped data
2) Assume the trained parameters $(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots, \boldsymbol{\theta}_s)$ are samples of the posterior $\boldsymbol{\theta}_i \sim P(\boldsymbol{\theta}|\mathcal{D})$



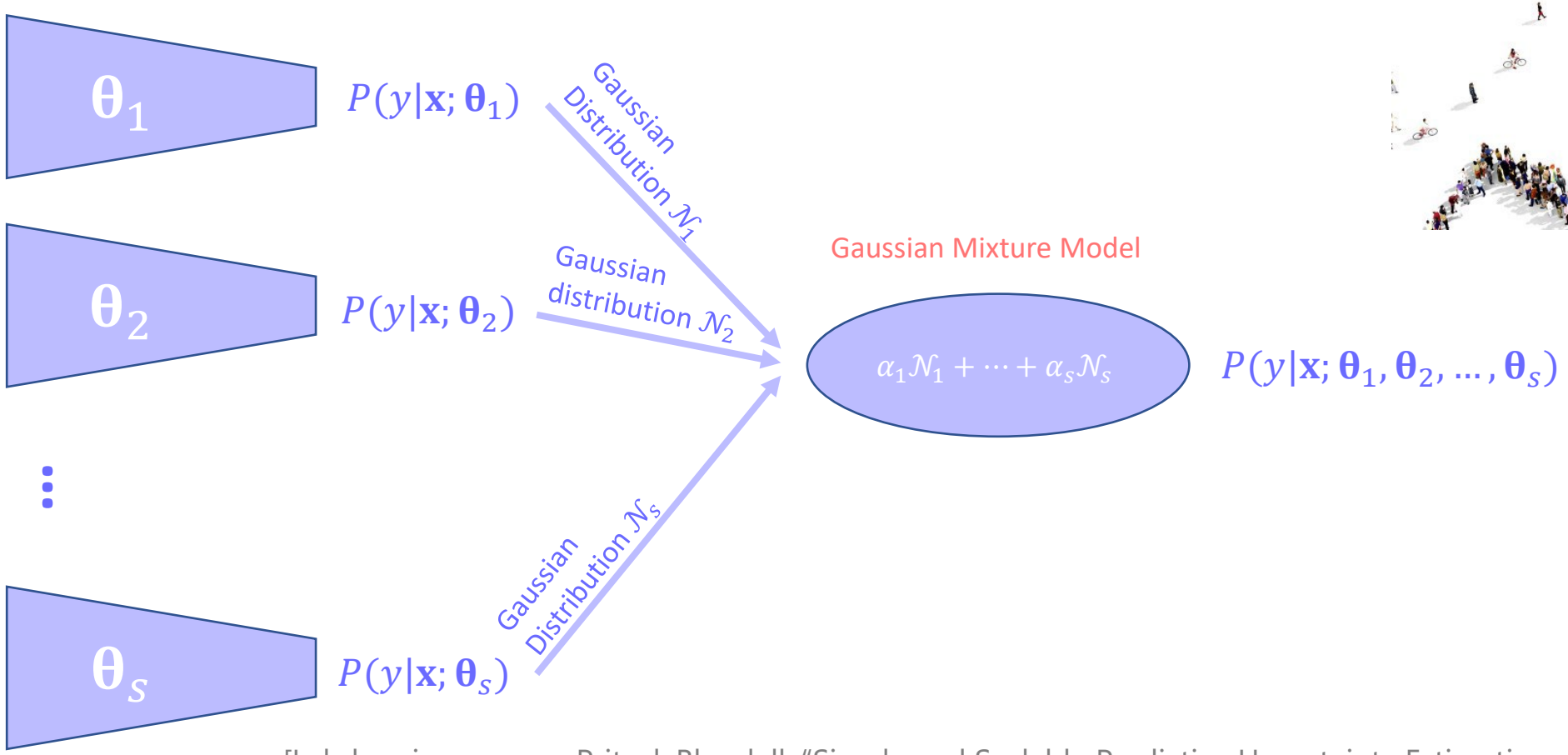
[Osband, Blundell, Pritzel, Van Roy, "Deep exploration via bootstrapped DQN", NIPS 2016]

Simple&Effective Techniques: Ensemble Methods

Ensemble-methods: procedure

1) Train s independent networks using different network initializations and bootstrapped data
2) Have each network model heteroscedastic uncertainty within MLE





[Lakshaminarayanan, Pritzel, Blundell, "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles", NIPS 2017]

Simple&Effective Techniques: Ensemble Methods

Ensemble-methods: important properties



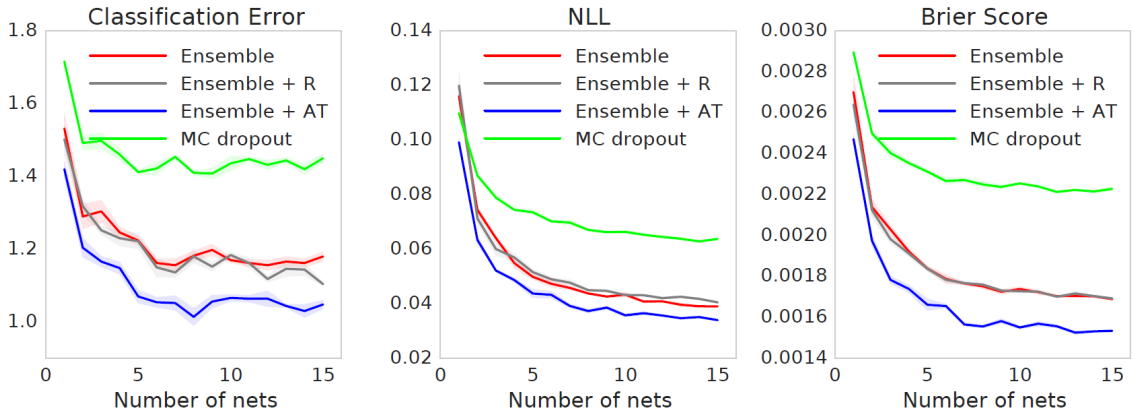
- Multiple point estimates of the posterior (frequentist approach)
  - Different initialization will give rise to different samples from the posterior
- Not much parameters to tune
- Model structure remains the same
- Computationally convenient training (parallelizable)
- Memory-wise not so efficient
- Evaluation not so efficient

[Lakshaminarayanan, Pritzel, Blundell, "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles", NIPS 2017]

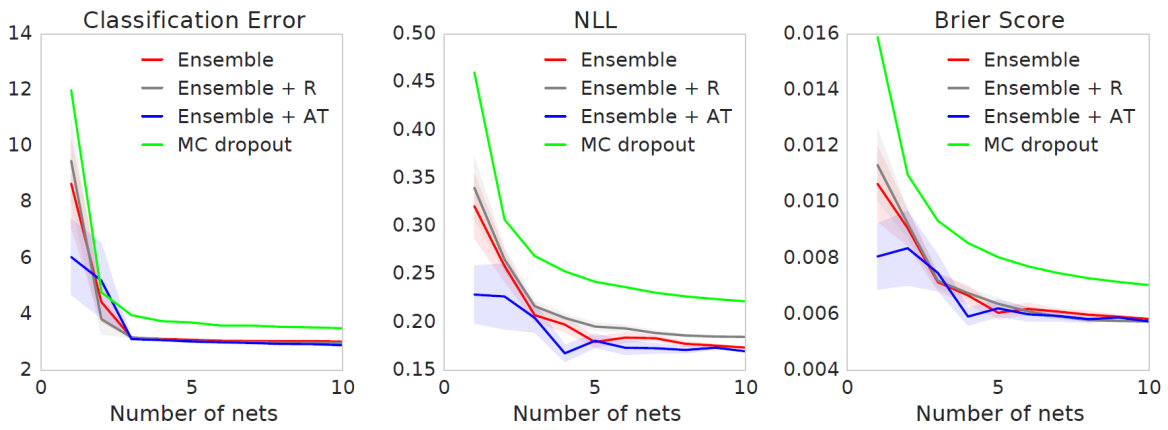
[Osband, Blundell, Pritzel, Van Roy, "Deep exploration via bootstrapped DQN", NIPS 2016]

Simple&Effective Techniques: Ensemble Methods

Ensemble-methods: Results



(a) MNIST dataset using 3-layer MLP

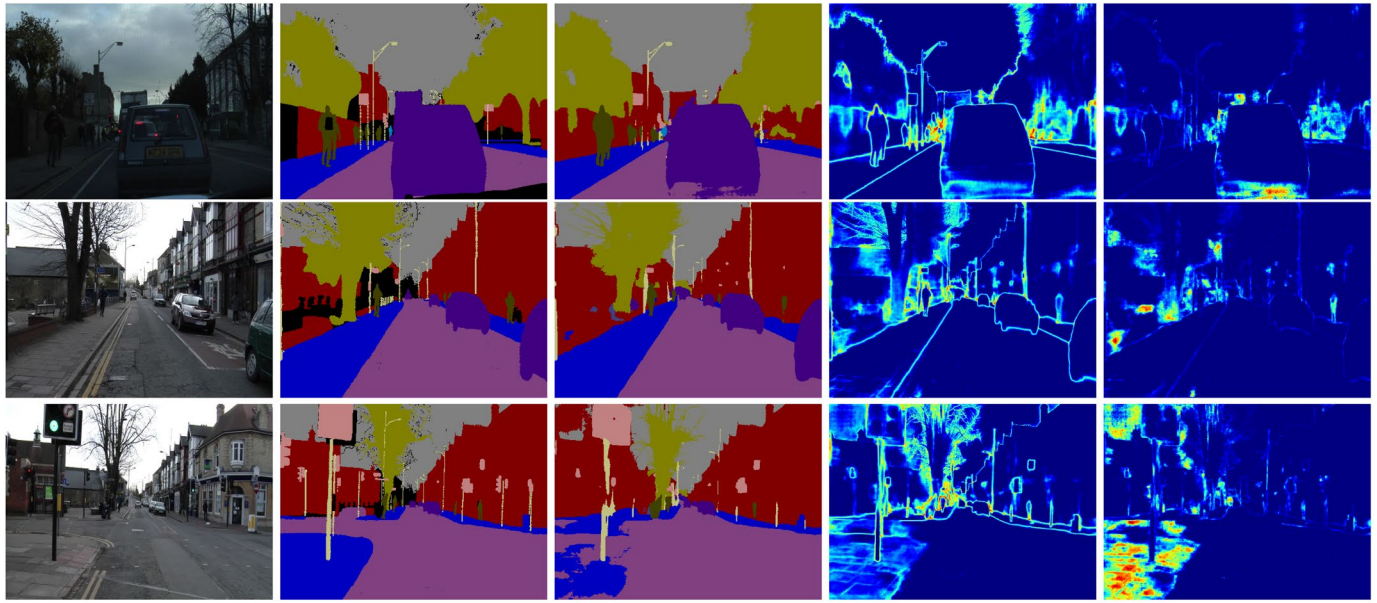


(b) SVHN using VGG-style convnet

[Lakshaminarayanan, Pritzel, Blundell, "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles", NIPS 2017]

Simple&Effective Techniques: Ensemble Methods

Ensemble-methods: + MCDO



(a) Input Image (b) Ground Truth (c) Semantic Segmentation (d) Aleatoric Uncertainty (e) Epistemic Uncertainty

[Kendall and Gal “What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?” NIPS 2017]

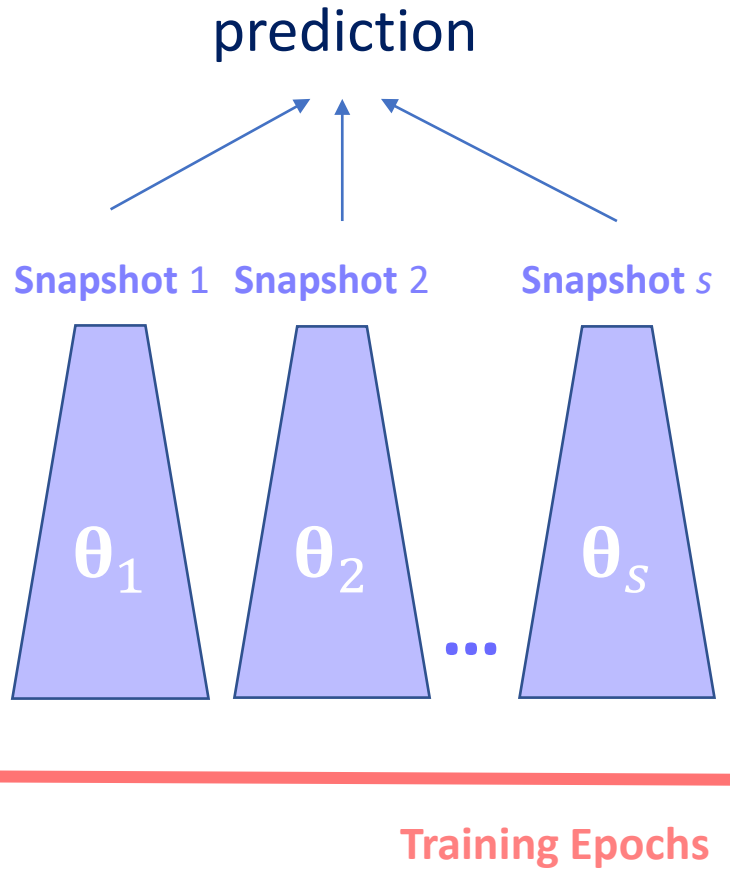
How to make Ensembling more efficient?



Simple&Effective Techniques: Efficient Ensemble

Efficient Ensemble: SWAG

optimization advances without using snapshots

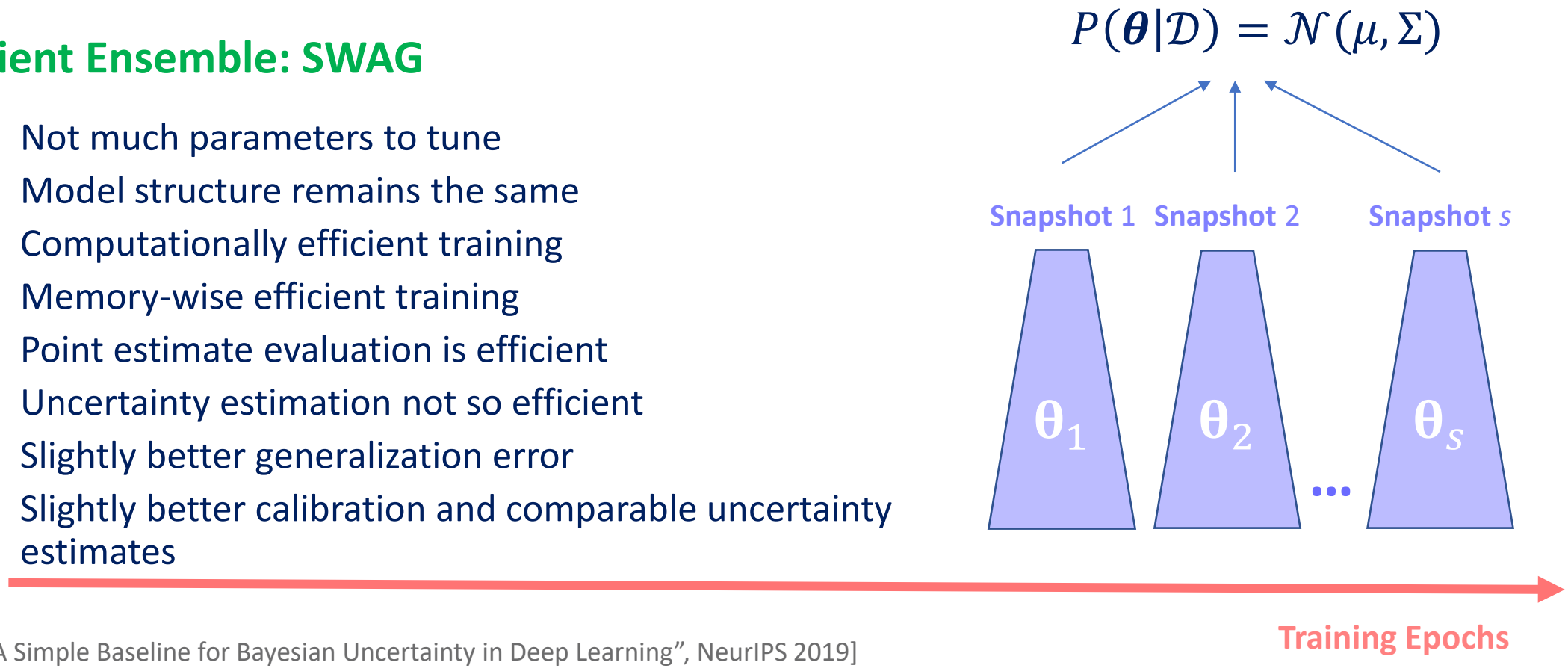


["A Simple Baseline for Bayesian Uncertainty in Deep Learning", NeurIPS 2019]
["Averaging Weights Leads to Wider Optima and Better Generalization", UAI 2018]
["Loss Surfaces, Mode Connectivity, and Fast Ensembling of DNNs", NeurIPS 2018]
["Improving Stability in Deep Reinforcement Learning with Weight Averaging", UAI UDL workshop 2018]
["There Are Many Consistent Explanations of Unlabeled Data: Why You Should Average", ICLR 2019]

Simple&Effective Techniques: Efficient Ensemble

Efficient Ensemble: SWAG

- Not much parameters to tune
- Model structure remains the same
- Computationally efficient training
- Memory-wise efficient training
- Point estimate evaluation is efficient
- Uncertainty estimation not so efficient
- Slightly better generalization error
- Slightly better calibration and comparable uncertainty estimates



["A Simple Baseline for Bayesian Uncertainty in Deep Learning", NeurIPS 2019]
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